

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2023/2024

SEPTEMBER EXAMINATION

UCCD3113 DISTRIBUTED COMPUTER SYSTEMS

THURSDAY, 12 OCTOBER 2023

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING

Instruction to Candidates:

This question paper consists of THREE (3) questions in Section A and TWO (2) questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Candidates are allowed to use non-scientific calculator.

Answer questions only in the answer booklet provided.

UCCD3113 DISTRIBUTED COMPUTER SYSTEMS**SECTION A**

- Q1. (a) Generate a smart plantation solution (architecture diagram) with Amazon Web Services (AWS) cloud platform. Justify the selection of each cloud service. Your design should consider:
- Deploying sensors for data capturing,
 - Storing data for plant growth analysis,
 - Generating serverless functions for data processing,
 - Notifying farmer for abnormal farm's conditions,
 - Having data pipeline for different cloud services,
 - Storing sensor log files into the cloud storage,
 - Visualizing sensor data for trend analysis, and
 - Integrating some relevant intelligent cloud services.
- (12 marks)
- (b) There are different sensors installed at various locations of an industrial plant, continuously gathering data from machines and devices. The data is crucial to be analyzed in real-time with a proper analytics tool, so that faults and failures can be resolved in minimum time. This is the core purpose of an industrial IoT system. Based on this scenario, which cloud deployment model would you recommend to the plant owner? Justify your selection with **TWO (2)** supporting reasons. (3 marks)
- (c) AWS offers several compute and database options to meet different needs. Which AWS service(s) can be used to host an off-premise MySQL database with minimal administrative operations? Justify your answer. (4 marks)
- (d) Answer the following questions with short justification in the context of AWS.
- (i) A company wants to utilize AWS Storage. For the company:
- low storage cost is paramount,
 - the data is rarely retrieved, and
 - data retrieval times of several hours are acceptable.
- What is the best AWS storage option to use? (2 marks)
- (ii) Provide **TWO (2)** features of Amazon QuickSight that aid in deriving business value from its data. (2 marks)
- (iii) An external audit is being conducted on UTAR. The IT auditor requires a log of all access to the AWS resources in UTAR's account. Which AWS cloud service can assist in providing these details? (2 marks)
- [Total : 25 marks]

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- Q2. (a) From your point of view, how does ChatGPT utilize cloud computing technology to enhance its natural language processing capabilities? Please provide **THREE (3)** points to support your justification. (6 marks)
- (b) Define microservices and present **TWO (2)** advantages of microservices over traditional web services. (5 marks)
- (c) In a distributed system scenario, under what circumstances would you choose message queuing over the publish/subscribe model? Please provide **THREE (3)** points to support your justification. (6 marks)
- (d) Compare and contrast the client-server architecture with the N-tier architecture in the context of distributed systems. Discuss the key differences (in a table) in their **design, functionalities, security, and scalability**. (8 marks)
- [Total : 25 marks]
- Q3. (a) Suppose there are four processes W , X , Y and Z . All clocks run at the same rate (1 unit of time) but initially W 's clock reads 12, X 's clock reads 0, Y 's clock reads 5 and Z 's clock reads 9. At time 12 by W 's clock, W sends a message to X , this message takes 4 units of time to reach X . X then waits one unit of time and then sends a message onto Y which takes 2 units of time to reach Y . Later, Y also waits one unit of time and then sends a message onto Z . The message reaches Z after 1 unit of time. After Z waits for 2 units of time, a message takes another 1 unit of time to reach back to Y . Assuming that the system implements Lamport's timestamps, draw a picture to illustrate the timestamps for the messages and how the timestamps are obtained. (8 marks)
- (b) Four clocks W , X , Y and Z are performing internal synchronization by using Berkeley algorithm. W is reading 3:00:00.0 (hr:min:sec) when it is chosen as the master clock. At the same time, X , Y and Z are reading 2:50:00.0, 3:18:00.0 and 3:24:00.0, respectively. Illustrate the clock synchronization process with diagrams from initialization to adjust the time and show the calculations. (10 marks)
- (c) Some networks, such as IPTV service, provide an asymmetric bandwidth. However, Cristian's algorithm assumes symmetric delays to and from the server. You are on a network with 6 Mbps downstream bandwidth and 4 Mbps upstream bandwidth, respectively. Your client makes a request at 8:30:00.0 and gets a response 400 msec later. If the time on the server is 7:32:00.0, what would the time be set to on your client? (4 marks)

UCCD3113 DISTRIBUTED COMPUTER SYSTEMS**Q3. (Continued)**

- (d) A clock, X , is reading 22:28:20.0 (hr:min:sec) when a clock synchronization request is sent to time server, S . After receiving the request from X , S prepares a response and appends the time 22:29:58.0 from its own clock. What will be the time set by X if the round-trip time (RTT) takes 8.0 seconds? Show your formula and calculation using Cristian's algorithm. (3 marks)

[Total : 25 marks]

SECTION B

- Q4. (a) Introduce **THREE (3)** key integration methods available for building distributed systems. (6 marks)

- (b) Construct a LAN diagram for a new shopping mall. Your diagram should consider providing internet services to the shopping mall internal staff and customers. Network components in the diagram should include (but not limited to) various types of servers, network switches (hub, router, gateway), and networking equipments (printer, kiosk, computer, mobile devices).

[TIPS: You should group the staff and customer into different subnets]

(11 marks)

- (c) Provide **THREE (3)** characteristics of cloud architecture that separate it from the traditional one. (6 marks)

- (d) What is a Lamport's logical clock? (2 marks)

[Total : 25 marks]

- Q5. (a) Discuss **THREE (3)** primary challenges in managing and maintaining a cluster computing system, and how are these challenges addressed? (9 marks)

- (b) How does grid computing contribute to scientific research and data-intensive applications, such as climate modeling or genomic analysis? Please provide **THREE (3)** points to support your justification. (6 marks)

- (c) How can peer-to-peer networks be used for content distribution, such as file sharing or video streaming? (4 marks)

UCCD3113 DISTRIBUTED COMPUTER SYSTEMS**Q5. (Continued)**

- (d) Normally, we need to configure AD, DNS and DHCP servers during private cloud setup. What are the functionalities of each component?
- (i) Active Directory (AD). (2 marks)
 - (ii) Domain Name System (DNS) server. (2 marks)
 - (iii) Dynamic Host Configuration Protocol (DHCP) server. (2 marks)

[Total : 25 marks]
